2025 IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN) | 979-8-3315-2042-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICMLCN64995.2025.11140457

Deep Q-Network Based UWB Anchor and Strategy Selection for Accurate Pedestrian Localization in Vehicular Environments

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***Abstract*—Ultra-wideband (UWB) technology is emerging as a key technology for indoor navigation and positioning systems, offering reliable, accurate, and energy-efficient performance. This characteristics of UWB can be leveraged to localize pedestrians in vehicles equipped with UWB technology. However, the localization of pedestrians in a vehicle environment, which is a more complex scenario than an indoor environment, requires further consideration. In particular, the selection of a line-of-sight (LoS) link and the delusion of pecision (DoP) problem are of paramount importance for the performance of the localization process. To address these issues, we propose a deep Q-network based UWB anchor and localization strategy selection that improves the localization success rate by identifying LoS links without relying on complex LoS/NLoS identification methods. Additionally, it determines the optimal localization method and addresses DoP issues to enhance location accuracy. The proposed scheme is evaluated against existing techniques in terms of localization precision at varying distances, LoS/NLoS conditions, and scenarios involving vehicle and pedestrian moving. The results demonstrate that the proposed approach outperforms existing methods.**

***Index Terms*—Ultra-wideband (UWB), Localization, Pedestrian, reinforcement learning, Deep Q-Network**

I. Introduction

The demand for positioning systems has increased significantly with the recent growth of Internet of Things (IoT) applications. Indoor positioning systems in particular use a variety of wireless technologies, each with its own strengths. Typical wireless positioning technologies used in indoor positioning systems include ZigBee, WiFi, radio frequency identification (RFID), Bluetooth Low Energy (BLE), and impulse radio ultra-wideband (IR-UWB). Of these, UWB is quickly becoming the dominant technology for indoor positioning systems [1]. UWB uses high-bandwidth communications, allowing the transmission of very short pulses in the time domain. This unique feature provides a significant advantage in distinguishing individual multipath elements. By effectively extracting precise temporal information from these signals, UWB enables highly accurate position estimation, making it particularly suitable for modern applications that require low latency and high reliability [2].

The growing use of UWB technology is having a notable influence on the development of products across a range of industries. For instance, well-known smartphone brands such as Apple and Samsung have integrated UWB chips into their products, while several leading automotive manufacturers have also introduced UWB-based digital key fob applications.

The integration of UWB technology in vehicles and smartphones has prompted a recent study PedLoc [3] to propose a solution for tracking the location of pedestrians carrying UWB devices via in-vehicle UWB. This study employs the IEEE 802.15.4z standard [4], particularly the double side two-way ranging (DS-TWR) approach, to develop an effective communication protocol optimized for the automotive environment. Additionally, the proposed scheme achieves improved pedestrian localization accuracy compared to traditional methods by employing channel quality metrics to identify the line-of-sight (LoS) link and selecting a set of anchors, including the root and candidate anchors.

However, if anchors with LoS link status are exclusively selected for localization, a DoP problem may ensue, as illustrated in Fig. 1. In the scenario depicted on the left, the estimated localization area (green area) is small. Conversely, in the scenario on the right, the estimated localization area is large, resulting in a substantial localization error. In other words, localization accuracy exhibits significant variations depending on the selected anchor's position and the tag's position [5]. This is particularly critical in vehicle environments, where the spacing between anchors is very limited. Consequently, when selecting anchors, greater attention must be paid to the position between the anchor and the tag.

Furthermore, mathematical anchor selection techniques using channel quality metrics have difficulty adapting to the dynamic nature of the vehicular environment, including buildings, traffic lights, utility poles, other vehicles, and trees.

To address the issue of not adapting to the dynamic nature of the environment, recent research in indoor localization over wireless networks has concentrated on applying deep reinforcement learning (DRL) to develop environmentally robust anchor selection techniques [6], [8]. The results of these

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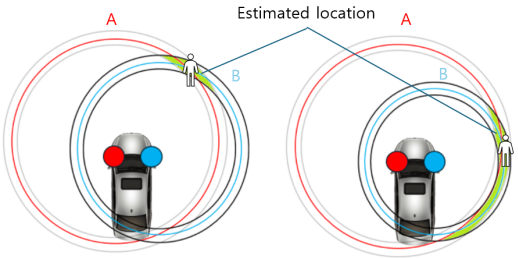

Fig. 1: Delusion of Precision problem on the vehicle.

studies demonstrate that the application of anchor selection via DRL yields superior performance in indoor location tracking compared to anchor selection via mathematical techniques. However, the application of DRL in vehicular environments presents unique considerations that are distinct from those encountered in indoor environments. Consequently, we propose a deep Q-Network based pedestrian localization (DQN-PedLoc) for UWB anchor and strategy selection that can effectively accommodate the complexity typical of vehicular environments.The contributions of this paper are presented below.

- The vehicle's complex environment can be defined as a Markov decision process (MDP), and the defined MDP model can be utilized to select a set of optimal LoS links from the current state. This approach facilitates more precise pedestrian localization in vehicles.
- Our methodology differs from other DQL approaches to UWB anchor selection in that it does not employ a fixed localization technique. Instead, the action space is divided into two categories, enabling the application of different localization techniques depending on the category to which the selected action belongs. The strategy selection design offers a flexible combination of localization techniques, providing a robust solution in complex vehicular environments and addressing the DoP issues.
- In contrast to previous indoor localization DQL approaches that have been tested in simulated environments, we constructed a test environment using real vehicles and UWB equipment to conduct our experiments. This approach provides more accurate and practical insights into the effectiveness of the proposed localization techniques in dynamic vehicle scenarios.

II. Related Works

In the field of indoor location tracking using conventional UWB, research studies are ongoing with the objective of determining LoS/NLoS links. According to these studies, using a small number of anchors from LoS links improves location accuracy compared to using all installed UWB anchors [10]. The authors of [7] have proposed a UWB node selection method based on the Mean Square Error (MSE) metric. This method involves dividing the set into several groups, with the one exhibiting the lowest MSE proceeding with location tracking using the weighted least squares method.

However, these mathematical approaches are difficult to adapt to different real-world environments, especially obstacles. For this reason, there has been a notable increase in recent research on the use of DRL for indoor localization [6], [8]. In [6], a framework is proposed that utilizes a Q-Network to perform anchor selection, taking into account the energy of UWB anchors and the existence of LoS links. The framework demonstrates high energy efficiency and accuracy. Building on these works, our DQN-PedLoc employs DRL as an anchor selection technique. However, [6] uses the ground truth value of the tag as the state of the MDP. This is only possible on a simulation basis and is difficult to apply in the real world because we don't know the exact location of the previous tag. In contrast to existing indoor localization DQL approaches, DQN-PedLoc, our proposed, establishes a test environment using real vehicles and UWB equipment to conduct experiments.

Another line of research addresses environmental issues by focusing on the UWB channel impulse response (CIR) [3], [9]. The authors of [9] present a method for training a deep neural network to classify the LoS/NLoS status of UWB signals. Although the study demonstrates high classification accuracy, it is contingent upon training data, which constrains its applicability to vehicular environments.

In [3], the channel quality is evaluated through the three peaks of the CIR, and the anchor with the highest channel quality is selected as the root anchor, and the anchor with the second highest channel quality is selected as the candidate anchor to localize the position. They also propose a new localization technique that combines Time of Flight(ToF) and Time Difference of Arrival(TDoA) methods to solve the DoP problem. However, the selected anchors through channel quality may still face the DoP problem. Additionally, CIR analysis can be impractical in an automotive environment where real-time is critical.

III. System Design

To identify a set of anchors situated in proximity to the LoS link and mitigate the potential issues associated with the DoP when localizing pedestrians from a vehicle, we propose a Deep Q-Learning based vehicle-mounted UWB anchor and strategy selection approach.

To enhance the efficiency of vehicle-to-pedestrian localization, the UWB communication protocol between vehicle and pedestrian(tag) in DQN-PedLoc employs a method similar to that described in reference [3]. All anchors installed on the vehicle are in listen mode. The pedestrian's tag transmits an advertising (ADV) message. Upon receipt of the aforementioned message, the primary anchor is to initiate the DS-TWR process with the tag by transmitting a POLL message. The remaining anchors, which were not selected as primary anchors (including secondary anchors), enter leaf mode and listen to the communication between the tag and the primary anchor. Once communication between the tag and the primary anchor is complete, the vehicle estimates the tag's location based on the communication results from the selected primary and secondary anchors. The selection of the set of primary and secondary anchors is based on the values with the highest probability in the Q-network, as determined by the current

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state. The following sections present a detailed construction of the MDP model and an account of the procedures for training the Q-network.

A. MDP Model

1) Agent: In contrast to other UWB deep Q-learning studies that have designed mobile users or tags as agents, the proposed framework positions the vehicle as the agent. Rather than undertaking actions related to the movement of the vehicle, the vehicle performs the actions of selecting the primary and secondary anchors. This is described in greater detail below.

2) Action-Space: In the proposed framework, the action space is defined as the selection of a primary anchor for communication with tags and secondary anchors for assisting in positioning. The aim of this study is to identify the optimal primary anchor for communication with the tag, as well as the most suitable secondary anchors from the available options, with the goal of maximizing localization accuracy.

The number of secondary anchors may vary from one to two. In the event that a single secondary anchor is present, the position is determined through the use of ToF acquired from the primary anchor and TDoA obtained from the secondary anchor. In the event that two secondary anchors are present, the ToF from the primary anchor is combined with the TDoA from the secondary anchors in order to estimate the ToF of the secondary anchor. Subsequently, the intersection of the three circles is employed to ascertain the location. The variety in the number of secondary anchors is designed to integrate the selection of localization strategy techniques into the learning process. The selected action, denoted by a_t, at time slot t, can be represented as either $a_t = [a_i, a_j]$ or $a_t = [a_i, a_j, a_k]$, where a_i represents the primary anchor and a_j and a_k represent the secondary anchors. The number of secondary anchors is indicative of the position calculation method itself and contributes to reducing the risk of DoP problems that can occur when selecting anchors on the same axis to measure position.

3) State-Space: The action is selected at time slot t based on the current state of the system s_t. Each state consists of the azimuth AZ_t that between the tag and the vehicle at the previous time $t-1$, the primary anchor P_t used at time $t-1$, and the reception results from the remaining anchors O_{tj} that overhear the communication between the primary anchor and the tag.

a) AZ_t: Azimuth between the tag and the vehicle at the previous time $t-1$. Our agent does not know the current location of the tag; It only knows the localization results obtained from previous UWB communications. The AZ_t is rounded to the nearest integer. so, AZ_t can take values between $0°$ and $359°$.

b) P_t: The primary anchor at the previous time $t-1$. This is entirely equivalent to a_i selected from a_{t-1} at the previous time $t-1$, for $1 \leq i \leq N_v$.

c) O_{tj}: The vector O_{tj} represents the reception times at which anchor j overheard to the DS-TWR message between anchor P_i and tag at the previous time $t-1$ for $(1 \leq j \leq i-1)$ and $(i+1 \leq j \leq N_v)$. Each element of the vector takes a value of either 0 or the reception times, where 0 indicates that the corresponding anchor j did not overheard that message. In our system, DS-TWR consists of four messages: POLL, RESP, FINAL, and ADD. So each $O_{t,j} = (o_{t,j,1}, o_{t,j,2}, o_{t,j,3}, o_{t,j,4}), \quad o_{t,j,k} \in \{0,1\}$ for $k = 1,2,3,4$. In order to allow our model the greatest flexibility to respond to a wide variety of scenarios, it is preferable to take the overall reception time for each of the four messages as the state variable, rather than simply determining if anchor j has received all of the messages.

4) Environment: In this framework, the environment is defined as everything that occurs outside the vehicle. The agent's role is limited to selecting a primary anchor for direct communication with the tag and secondary anchors for position measurement based on its current state. The movement of the pedestrian, the UWB communication between the selected primary anchor and the tag, the reception status of the remaining anchors, and the reward r_t given to the agent from the position estimation results represent the interaction between the environment and the agent.

a) Pedestrian: The pedestrian holding the tag is situated in an environment that is beyond the scope of our control, resulting in a random trajectory. To represent this stochastic movement as a state, we calculate the azimuthal angle between the tag and the vehicle. With the azimuth, regardless of the pedestrian's random movements, we can represent it as AZ_t, irrespective of distance.

b) UWB communication: Based on the previous UWB communication, the state is passed as input to the Q-network of the framework, which generates a primary and secondary anchor as output. The agent then performs UWB communication with the tag through the primary anchor, thereby enabling it to obtain the ToF values between the primary anchor and the tag and to estimate the tag's location using the values received from the secondary anchors.

5) Reward: : In our DQN model, the reward function is designed to encourage the agent to select a primary anchor that ensures effective communication with the tag and to choose secondary anchors that improve localization accuracy. When an action is chosen and communication with the tag is carried out, the resulting data is used to calculate the estimated azimuth angle of the user. This estimated azimuth angle is then compared to the ground truth GPS value and represented as E_t, is calculated as

$$E_t = \left(\frac{180}{\pi}\right)^2 \cdot \left|\tan^{-1}\left(\frac{x_t - x_v}{y_t - y_v}\right) - \tan^{-1}\left(\frac{x_{gt} - x_v}{y_{gt} - y_v}\right)\right|^2 \quad (1)$$

where (x_t, y_t) is the estimated position of Tag at time slot t and (x_v, y_v) is the vehicle's position(GPS) and (x_{gt}, y_{gt}) denotes the ground truth of tag(GPS). Our model will select action that minimize E_t. At the same time, we need to check whether the selected secondary anchors is well chosen.

$$E_{bset,t} = \min E_{i,t}, \quad i \in I \quad (2)$$

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Find the smallest $E_{best,t}$ among all possible secondary anchors $i \in I$. Let I be the set of all possible sets containing the secondary anchor that agent have chosen at the time t.

The reward function $R(s_t, a_t)$ is defined as

$$R(s_t, a_t) = \begin{cases} 10, & E_{best,t} < \delta, E_t < \delta \\ 1/E_t, & E_{best,t} < \delta, E_t > \delta \\ -10, & E_{best,t} > \delta \end{cases} \tag{3}$$

where δ is target threshold value for allowable angular error. $E_{bset,t}$ and E_t. If both $E_{bset,t}$ and E_t are above the threshold δ, we consider that we have chosen the primary and secondary anchors well, and give the largest reward. If $E_{bset,t}$ crosses the thread hold δ, but E_t does not, then only the primary anchor is considered a good choice, and a relatively low reward is given. If $E_{bset,t}$ itself does not cross the threshold δ, it is judged that the primary anchor itself was not chosen well, and a negative reward is given.

B. Deep Q-Network for Pedestrian Localization

In scenarios comprising a substantial number of state-action pairs, the computational expense associated with traditional Q-learning algorithms is considerable. Accordingly, in contrast to the traditional approach of computing a Q-table, we employ a deep neural network (DNN) to select actions based on the agent's current state through the use of a deep Q-network. In a DQN, the Q values obtained from traditional Q-learning are treated as optimal values, and a function approximator is used to estimate those Q values: $Q(s_t, a_t; \theta) \approx Q^*(s_t, a_t)$; Furthermore, since it can be challenging to use histories of arbitrary length as input to the neural network, the state is based on a fixed-length representation generated by a state reformulation ϕ. The detailed DQN training scheme is presented in Algorithm 1.

Initiating each episode, an arbitrary action is selected to establish the initial state $s1$, after which UWB communication is performed in accordance with the selected action. During the course of each episode, an ϵ-greedy policy is employed to determine the action to be taken at each step. To maintain a balance between exploration and exploitation in our framework, we employ an ϵ-greedy action selection policy. The maximum value of ϵ, denoted by ϵ_{start} is set to 1.0, thereby establishing a baseline from which the ϵ value subsequently decreases. The ϵ continues to decrease by $\Delta\epsilon = \epsilon_{end} + (\epsilon_{start} - \epsilon_{end}) \cdot \exp\left(-\frac{steps}{\epsilon_{decay}}\right)$ until it reaches $\epsilon_{end} = 0.01$, at which point it stops decreasing.

When the greedy policy is applied, an action is obtained through the policy network Q_{policy}. Subsequently, the agent progresses UWB communication through the selected action, utilizing the communication value to determine the next state s_{t+1} and estimate the position of the tag. The reward r_t for the current step t is obtained based on the estimated position of the tag and the GPS value through the implementation of Equation 3. The state ϕs_t, action a_t, reward r_t, and next state ϕs_{t+1} obtained at time t are stored in the replay memory D. Then, a minibatch is randomly sampled from D. The sampled experience is used to update the policy network Q_{policy}, where a target network Q_{target} is used to compute y_i from the Bellman equation to stabilize the learning. The target network Q_{target} is updated using soft update at fixed intervals.

Algorithm 1 The DQNPL training algorithm

Initialize replay memory D, the policy network Q_{policy}, the target network Q_{target};
for each episode T **do**
  Observe UWB communication with a random selected action a_0 and acquire s_1
  Generate state reformulation $\phi(s_1)$
  for each step $t = 1 \rightarrow S$ **do**
    With probability ϵ select a random action at a_t
    Otherwise select action $a_t = \max_a Q_{\text{policy}}(\phi_{s_t}, a_t; \theta)$
    Observe UWB communication with action a_t and next state s_{t+1} and reward r_t
    generate state reformulation $\phi(s_{t+1})$
    Store the experience $\{\phi_t, a_t, r_t, \phi_{t+1}\}$ in the replay memory $\mathcal{D}$
    Sample a mini-batch $\{(\phi_i, a_i, r_i, \phi_{i+1})\}$ from the replay memory D
    for each experience $(\phi_i, a_i, r_i, \phi_{i+1})$ **do**
      Compute the target value:

$$y_i = r_i + \gamma \max_{a'} Q_{\text{target}}(\phi_{s_{i+1}}, a'; \theta^-)$$

      Update the policy network by minimizing the loss:

$$L = \sum_i (y_i - Q_{\text{policy}}(\phi_{s_i}, a_i; \theta))^2$$

    if step $t \ \% \ C = 0$ **then**
      Update the target network with Soft Update:

$$Q_{\text{target}} \leftarrow \tau Q_{\text{policy}} + (1 - \tau) Q_{\text{target}}$$

IV. Implementation and Evaluation

To assess the efficacy of the proposed DQN-PedLoc, we constructed a testbed comprising genuine vehicles and commercial UWB chips. Subsequently, we gathered training data to train the Deep Q-Network using the testbed. In this section, we will present the experimental setup and scenarios for train and test data, and provide a detailed evaluation of the results.

A. Experimental Setup

The Anchors and Tag utilized Qorvo's DWM3001CDK UWB board and mounted 6 anchors in a 2015 Kia Pride vehicle in accordance with the arrangement depicted in Fig. 2. The 6 anchors mounted in the vehicle were linked to a Galaxy Tab S4, which served as the central controller. The Tag links with the Galaxy S21 to obtain the pedestrian's GPS value. The ground truth values for all experiments are obtained from the GPS data of the Galaxy Tab S4 connected to the vehicle and the Galaxy S21 linked to the tag.

B. Experimental Scenarios

To train the Deep Q-Network of our DQN-PedLoc and evaluate it, we designed both static and dynamic scenarios. The Deep Q-Network was trained with data obtained through the training scenario and compared with related studies in simple scenarios. This comparison allowed us to evaluate the core components of DQN-PedLoc. Furthermore, the performance of DQN-PedLoc was evaluated in a realistic environment

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(a) Attached to the vehicle (b) Top view

Fig. 2: Vehicle-mounted 6 UWB anchors for TestBed.

through a dynamic scenario involving both vehicles and pedestrians.

Scenario	Distance(m)	Pedestrian	Vehicle	Link
Training	5-6	moving	stationary	Los
Case 1	10,20	stationary	stationary	Los
Case 2	3-25	moving	stationary	Los
Case 3	3-30	moving	moving	NLos

TABLE I: Scenarios Table

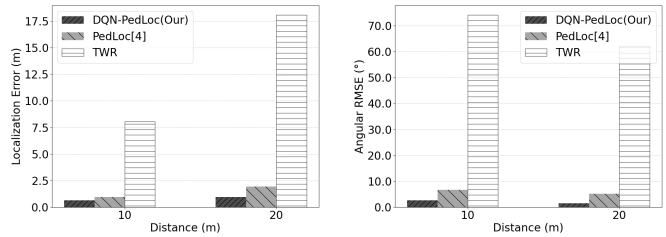
The experimental scenarios are presented in Table. I. All experiments were conducted in an outdoor parking area. The parking area was approximately 40% occupied at the time the experiments were conducted. The scenarios, which are relatively simple in nature (Training, Case1, and Case2), feature a vehicle and a pedestrian situated within an environment that allows for unobstructed line-of-sight (Los) between them. The vehicle is fixed at a specific location in each scenario. while the pedestrian with the tag is either stationary or moving.

In the training scenario, the pedestrian traverses a circular path around the vehicle, maintaining a distance of approximately 5-6 meters between them. At this time, the primary anchor is selected on a cyclical basis for the purpose of data collection. To facilitate comparison with related studies, two simple experiments, designated Case 1 and Case 2, have been conducted. In Case 1, the tag is located at 10, 20 meters from the vehicle, and the accuracy of each protocol is evaluated. In Case 2, we consider a situation where a pedestrian with a tag moves further and further away from the vehicle. The pedestrian initiates at a distance of 3 meters from the vehicle and subsequently moves away until they are approximately 25 meters away.

In Case 3, a more diverse environment is constructed, with both vehicle and pedestrian moving. The initial separation between the vehicle and pedestrian is greater than 30 meters, and they subsequently follow their respective trajectories. Additionally, throughout this scenario, most of the anchors remain in an NLoS condition, primarily due to parked cars in the parking area, which is approximately 40% full, as well as trees and other obstructions.

C. Experiment results

Fig. 3 illustrates the outcomes of Case 1. Fig. 3a illustrates the mean localization error for DQN-PedLoc, PedLoc, and TWR when the vehicle is stationary and the tag is situated at 10 and 20 meters. At a distance of 10 meters, the position error was found to be 0.65 meters for DQN-PedLoc, 0.97 meters for PedLoc, and 8.07 meters for TWR. At a distance of 20 meters, the position error was found to be 0.95 meters for DQN-PedLoc, 1.95 meters for PedLoc, and 18.07 meters for TWR. The difference between DQN-PedLoc and PedLoc is minimal. In contrast, there is a considerable difference between TWR and the other methods.



(a) Localization accuracy (b) Azimuth accuracy

Fig. 3: Comparing the localization and azimuth accuracy of DQN-PedLoc, PedLoc, and TWR in Case 1

Fig. 3b illustrates the angular accuracy at each location, as quantified by the Root Mean Square Error (RMSE). At a distance of 10 meters, the RMSE for angular measurements was found to be 2.69° for DQN-PedLoc, 6.68° for PedLoc, and 74.13° for TWR. At a distance of 20 meters, the position error was observed to be 1.54° for DQN-PedLoc, 5.14° for PedLoc, and 62.16° for TWR. As the distance increases, it becomes evident that the rate of growth in position error is greater than that of angle error. This indicates that we are more effectively addressing the DoP challenges associated with PedLoc, which is in line with our objective. In general, the results demonstrate that DQN-PedLoc exhibits superior performance, even exceeding that of PedLoc.

It is also noteworthy that there is a contrast between the training scenario and case 1. The training scenario is for distances of 5-6 meters, while case 1 is for distances of 10 and 20 meters. The efficacy of our DQN model in the training scenario and other environments can be attributed to the fact that we did not designate the distance to the tag as a state variable, a common practice among existing DQN models for anchor selection. Instead, we defined the azimuth and overhearing data as state variables. The advantage of defining the azimuth as a state variable is that we do not have to train on all distances, a strategy that enhances environmental robustness.

Fig. 4 illustrates the outcomes of Case 2. Fig. 4a illustrates the disparate outcomes of the tag's localization when the pedestrian is moving. It is observed that the results of DQN-PedLoc and PedLoc are predominantly situated in proximity to the trajectory of the tag, whereas TWR exhibits a pronounced variation. Fig. 4b depicts the localization error of the three algorithms relative to the tag trajectory. The localization error was found to be 0.3926 meters for DQN-PedLoc, 0.4151 meters for PedLoc, and 15.8551 meters for TWR. It can be observed that the aforementioned errors are not as reliable as those observed in Case 1, due to the presence of minor inaccuracies in the GPS values recorded by the Tag. However, the

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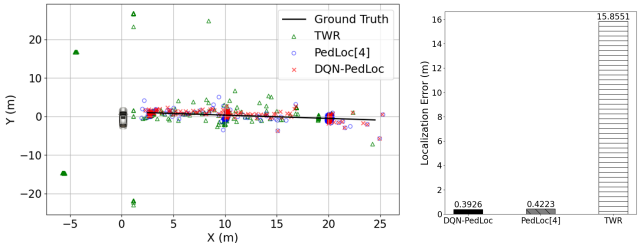
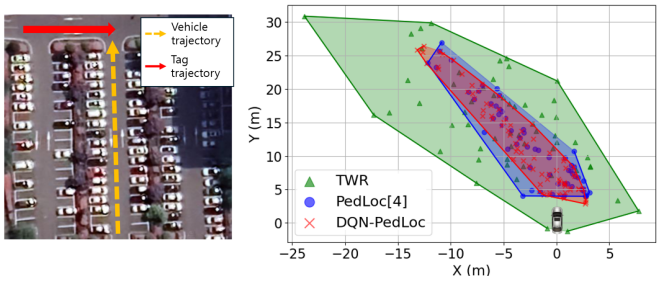


(a) Visualization result of the localization of each technology as the tag moves

(b) Location accuracy compared to ground truth

Fig. 4: Comparing the localization accuracy and visualization of DQN-PedLoc, PedLoc, and TWR in Case 2

similarities between these results and those obtained in Case 1 indicate that our DQN-PedLoc is effective in tracking the Tag's movement, irrespective of whether the Tag is stationary or moving.



(a) Each trajectory of the vehicle and pedestrian

(b) Visualization result of the localization of each technology

Fig. 5: Comparing the visualization of DQN-PedLoc, PedLoc, and TWR in Case 3

Fig. 5 presents a comparative analysis of the localization outcomes of the DQN-PedLoc and the comparator techniques in a dynamic scenario. Fig. 5a illustrates the respective trajectories of the vehicle and Tag in Case 3. The tag is moving in a rightward direction from its initial position, while the vehicle is moving in an upward direction from its starting point. For a certain distance, the position of the pedestrian within the vehicle is not visible and NLoS is maintained. Given the inherent difficulty in obtaining precise ground truth in a dynamic environment where both the vehicle and pedestrian are moving, we have elected to refrain from conducting a numerical analysis of the results. Instead, we have opted to present performance trends. Fig. 5b illustrates the relative position of the pedestrian in relation to the vehicle. Given that the vehicle is moving in an upward direction and the pedestrian is moving in a rightward direction, the position of the pedestrian as observed from the vehicle undergoes a shift from the top left to the bottom right. It should be noted that all techniques have variance due to the movement of both the vehicle and pedestrian. However, DQN-PedLoc shows relatively stable results, tending to follow a linear distribution down the diagonal. PedLoc demonstrates a similar trend to DQN-PedLoc, but some locations are far apart from the distribution. TWR has an irregular distribution and is relatively inaccurate.

V. Conclusion

In the context of UWB-based pedestrian localization installed in vehicles, this paper proposes a UWB node selection and subsequent localization selection technique based on DQN to achieve high localization performance. The DQN-PedLoc is trained to determine the optimal set of primary UWB and secondary anchors for pedestrian localization based on the UWB communication protocol proposed by [3]. The proposed DQN-PedLoc was evaluated in terms of position error and angle error with PedLoc, TWR under different diverse conditions, LoS/NLoS conditions, and vehicle and pedestrian movement conditions that differed from those present in the training data. The DQN-PedLoc demonstrated superior performance compared to the existing TWR, and also exhibited favorable outcomes when benchmarked against PedLoc. This demonstrates that our objective of establishing a more resilient MDP model in a vehicular setting was well-founded, and that we have resolved some of the DoP issues that PedLoc was unable to fully resolve. Moreover, in contrast to the conventional DQN-based indoor localization techniques, the real-world vehicle-based experiments have substantiated the effectiveness of UWB localization over DQN in actual vehicles.

Subsequent experiments will be conducted in a variety of environments, including those with randomly placed obstacles and multiple pedestrians, with the objective of localizing pedestrians in more realistic situations.

Acknowledgment

This work was partly supported by the National Research Foundation of Korea (NRF) grant funded by the BK21 FOUR program (NRF5199991514504).

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